

Comparing rates of primary medication nonadherence among patients at Vanderbilt Specialty Pharmacy to external specialty pharmacies

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PURPOSE

Compare primary medication nonadherence (PMN) rates between a health-system specialty pharmacy (HSSP) - Vanderbilt Specialty Pharmacy- and external specialty pharmacies (non-HSSP)

METHODS

- Study Design**
- Retrospective cohort study
 - Fill data obtained from HSSP software and Surescripts® (Non-HSSP patients)
- Setting**
- Vanderbilt Health System Oncology, Inflammatory (Rheumatoid Arthritis [RA], Inflammatory Bowel Disease [IBD], and Dermatology), or Multiple Sclerosis (MS) clinics
 - Pharmacy was categorized into HSSP and non-HSSP
- Sample**
- Inclusion Criteria:** Patients with a new specialty medication prescription between June 1, 2021, to May 31, 2022
 - Exclusion Criteria:**
 - Prescription/fill of a specialty medication within 180 days of the index prescription occurred
 - Alternative filling method used (inpatient, clinical trial, manufacturer free drug)
 - The ultimate outcome of the prescription could not be determined (e.g., patient lost to follow up or prescription sent to non-HSSP pharmacy with no confirmation of outcome)
- Analysis**
- Wilcoxon rank sum test: PMN rates by filling pharmacy and clinic type
 - Logistic regression model controlling for age, sex, clinic, race, insurance and filling pharmacy tested for associations with PMN

Lookback Window: 180 Days

Minimum duration prior to index prescription where patient may revert to naïve status and be considered as a "new" prescription and potential case of PMN.

Duplicate Window: 30 Days

Duration of time within which two sequential prescriptions for the same medication or therapeutic equivalent can be considered a duplicate.

HIGHLIGHTS

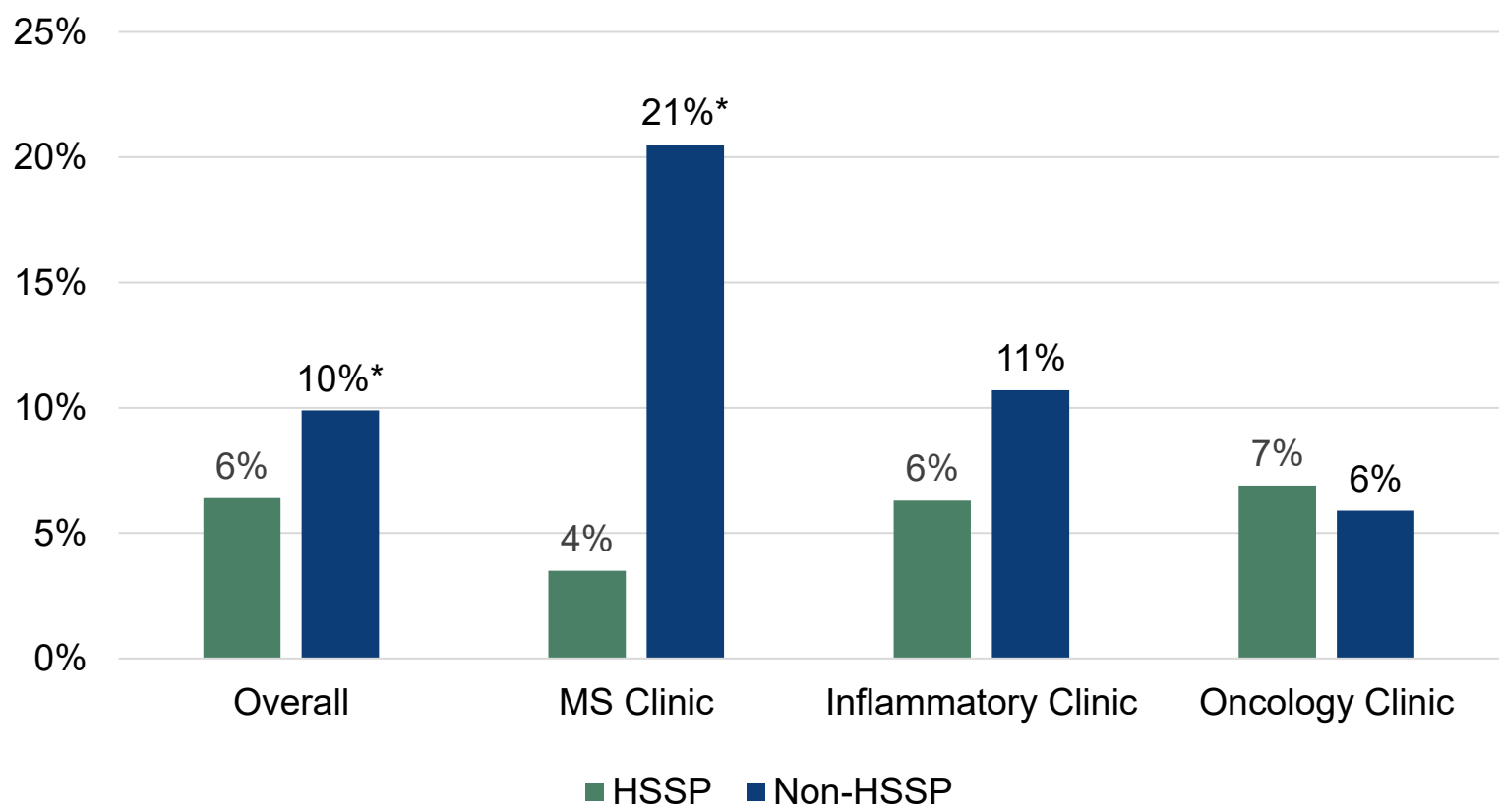
- Patients with a prescription sent to **non-HSSP pharmacies** were **60% more likely to have PMN**, indicating that HSSPs can help more patients access and initiate therapy compared to non-HSSPs.
- The reason for PMN differed by where the prescription was sent, with more patients **not obtaining insurance coverage if their prescription was sent to a non-HSSP pharmacy** and patient decision driving PMN for HSSP patients.

RESULTS

Table 1. Baseline Demographics

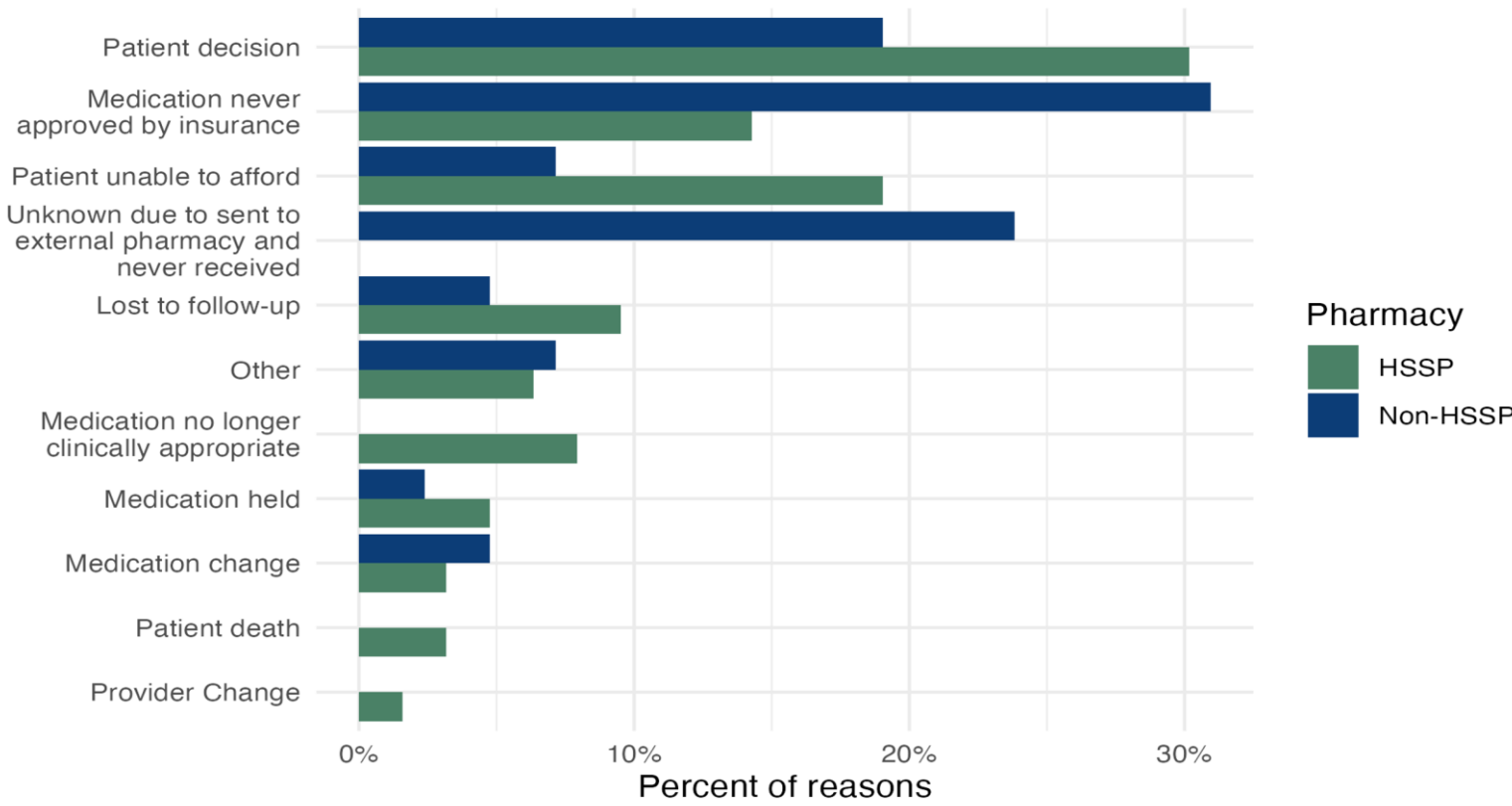
Characteristic	HSSP (N=987)	Non-HSSP (N=480)	Total (N=1467)
Age, years, median (IQR)	60 (45-70)	50 (35-61)	56 (41-68)
Female, n (%)	516 (52.3%)	295 (61.5%)	811 (55.3%)
Insurance, n (%)			
Commercial	495 (51.4%)	327 (69.6%)	822 (57.4%)
Government/Other	468 (48.6%)	143 (30.4%)	611 (42.6%)
Clinic, n (%)			
Multiple Sclerosis	58 (5.88%)	48 (10.0%)	106 (7.23%)
Inflammatory	306 (31.0%)	268 (55.8%)	574 (39.1%)
Oncology	623 (63.1%)	164 (34.2%)	787 (53.6%)

Figure 1. PMN Rates



*There was a significant difference in PMN rates between HSSPs and non-HSSPs for the **overall population (p=0.027)** and for patients in the **MS Clinic (p=0.013)**.

Figure 2. Reason for PMN



The most common reason of PMN for HSSP patients was **patient decision**. Non-HSSP patients were more likely to have PMN due to **medication never being approved**. The final outcome of prescription access was unknown for many non-HSSP patients.

Figure 3. PMN Identification and Results

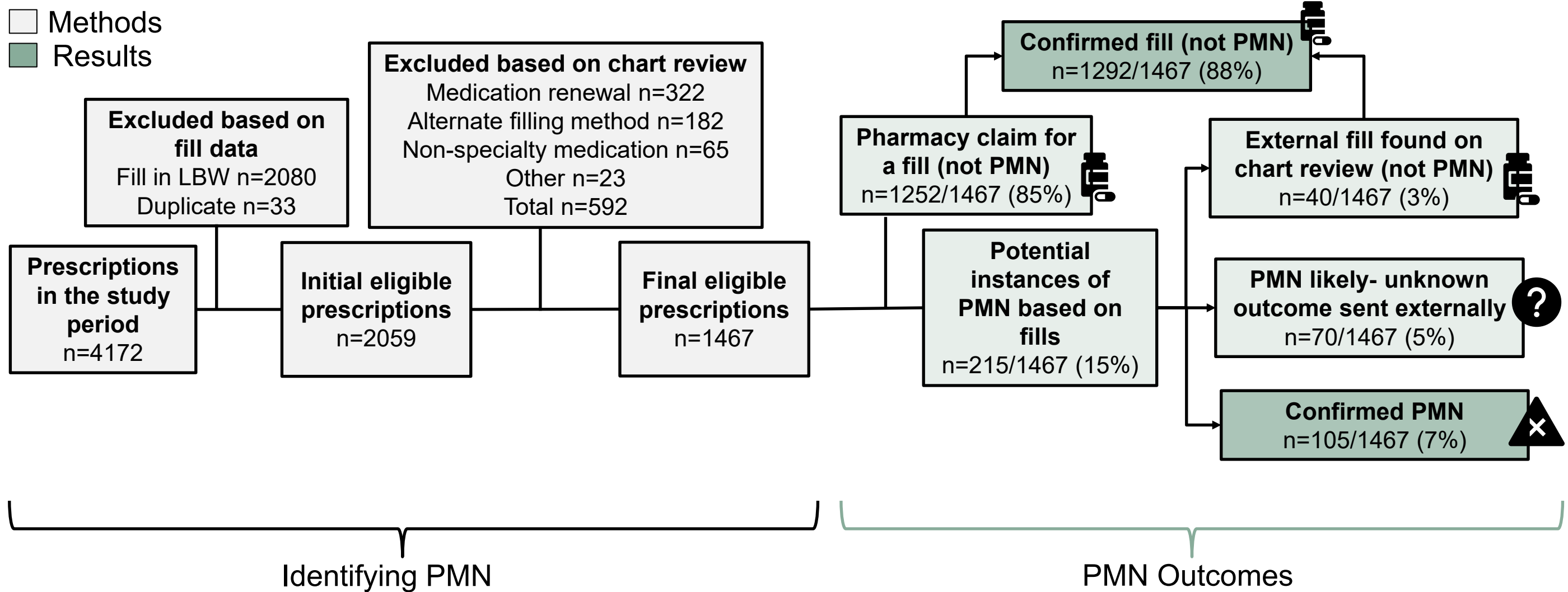
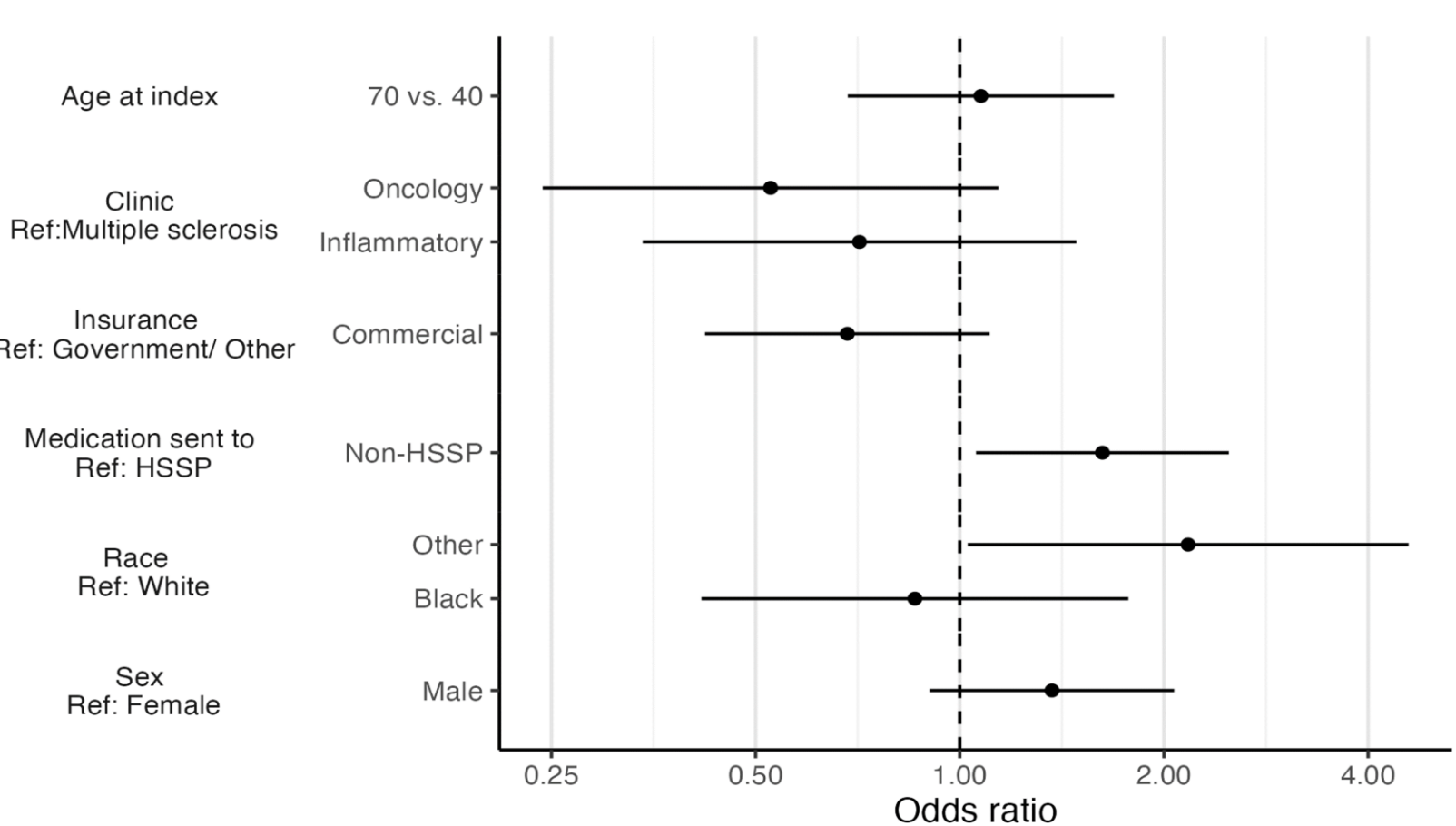


Figure 4. Logistic Regression Results



Non-HSSP patients were **60%** more likely to experience PMN